

CellSim

Species Stat Line Currency and Predictive AI Meeting Brief

Decision brief for the project planning meeting during the week of 21 September 2026

STATUS

Proposed for planning decision

OWNER

CellSim project

LAST UPDATED

17 September 2026

Prepared for	Josh and Sim
Evidence baseline	Current ProjectMain working-tree planning and implementation state reviewed 17 September 2026
Related records	Roadmap, Sprint 3 control record, Stat Line contracts, Scientific Data Economy, and Predictive AI research plan
Decision sought	Approve the operating boundary, near-term prerequisites, sequencing, and ownership for shared evidence work

1 Executive summary

Recommendation: keep the Species Stat Line, scientific-data currency, and Predictive AI on one evidence spine but in separate authority layers. The Stat Line records what happened. A versioned currency policy decides what selected observations are worth. Predictive AI forecasts and explains outcomes. Josh and Sim retain the authority to decide what is balanced, understandable, and ready for production.

This boundary lets the three efforts reinforce one another without allowing a payout formula or an AI score to redefine the simulation evidence. It also matches the current roadmap: Sprint 3 is closing the expedition contract and readability gaps; build and species calibration follow in Sprint 4; player comprehension follows in Sprint 5; production settlement and the first persistent Genome unlock remain Sprint 6 work. Predictive AI feature F20 stays on hold until the core loop, economy, and metric meaning are stable. [1][2]

GOALS AND NON GOALS

Goals	Non goals
Keep one trustworthy source for simulation facts and metric validity.	Choose a final payout formula in this meeting.
Make currency valuation versioned, replayable, and replaceable without changing historical reports.	Turn APS, AHS, RFS, or final population into an overall species rating.
Give upgrade authors a matched-seed evidence loop for prediction, design, and balance.	Reopen P3 or claim generalized predictive capability from bounded experiments.
Define the gate for one future P4 design-translation experiment.	Implement production Genome effects, costs, settlement, or persistence ahead of their roadmap gates.

DECISIONS REQUESTED

1. Adopt the four-layer boundary: measurement, valuation, prediction, and human decision.
2. Authorize a small contract package before economy tuning: metric definitions, a currency-policy-v1 interface, a predator dictionary, and a historical EX-007 erratum.
3. Keep survivor-count currency as prototype plumbing only until a Forest Edge reference panel is approved and candidate policies are compared offline.
4. Keep F20 paused during Sprint 3; after the contract gate, either select one bounded P4 translation question or explicitly continue the pause.

2 Roadmap position

The work is technically connected but scheduled across different gates. The measurement foundation already exists. The economic meaning, player comprehension, and persistent Genome loop do not. Treating those as separate readiness states prevents an implemented field from being mistaken for an approved game rule.

Workstream	Current state	Role in this brief	Next gate
Species Stat Line	Implemented role-specific calculation and export paths	Measurement authority	Freeze the metrics used by currency and forecasts
Scientific-data currency	Approved direction; survivor-count reward is prototype	Human-authored valuation policy	Approve reward and settlement contract
Upgrade design and balance	Mutation foundation exists; broader promotion review remains	Uses matched evidence to author tradeoffs and costs	Forest Edge reference panel and S4 co-design
Predictive AI	P0-P3 bounded complete; P4-P6 not started	Advisory forecast and diagnosis	One new human-owned P4 contract
Genome progression	Identity, profile snapshot, catalog, and UI foundation exist	Later consumer of approved settlement and balance	S6 persistent unlock and activation

WHAT IS ALREADY IMPLEMENTED

- Herbivore and predator Stat Lines use shared domain calculations for game display and export; phase and final windows are separate in the continued-world runtime. [4][5]
- The run result carries player population and currency earned, and the preview applies that value at phase boundaries and terminal completion. [10]
- The current factory sets currency earned equal to the surviving player population. This proves the reward path, not a validated economy. [3][10]
- The Genome identity, per-species profile, immutable launch snapshot, catalog, and Gene Lab display foundation are already recorded as complete. Production content, actions, costs, and persistence remain later work. [1][2]

READINESS JUDGMENT

The project is ready to define and test policy contracts against existing reports. It is not ready to lock final earning rates, use composite Stat Line scores as payout multipliers, or promote Predictive AI into production design authority.

3 Stat Line and currency

The Stat Line is most valuable when it remains descriptive. It tells designers what happened, over which window, with which inputs, and whether a derived value is valid. Currency is prescriptive: it decides which observations deserve reward and how much. The policy can read immutable Stat Line evidence, but it must not change metric meaning or overwrite a report.

Layer	Question answered	Owned output	Must not become
Raw evidence	What events and counts occurred?	Deaths, births, encounters, attempts, populations	A reward decision
Derived diagnostics	How should related facts be summarized?	Ratios, composites, and validity states	A universal quality score
Currency policy	What player behavior is worth data?	Versioned payout with caps and settlement	A second metric definition
Human review	Is the outcome balanced and fun?	Accept, revise, reject, or defer	An automatic consequence of one number

MINIMUM CURRENCY POLICY CONTRACT

A small currency-policy-v1 record should make tuning explicit and reproducible. It should contain:

- A policy ID and version tied to the ruleset, scenario, metric dictionary, and window.
- Exact metric IDs with validity, not-applicable, and role handling.
- Reward categories, thresholds, caps, diminishing returns, and discovery rules.
- Unique phase and final settlement IDs so repeated views cannot pay twice.
- A dry-run output from retained reports with no progression or simulation write.

CURRENCY DESIGN GUARDRAILS

- Do not pay raw tick count or uncapped common events.
- Do not treat final population as universally better; it needs a target band.
- Do not make APS, AHS, or RFS a direct multiplier.
- Compare candidate policies offline on the same retained seed panel.

KNOWN SEMANTIC DEBT

The current machine-readable dictionary defines RFS as Replication fitness score and APS as Actual prey score. The historical EX-007 report instead labels them Resource-finding score and Activity per step. Its accepted direct-outcome findings remain bounded evidence, but those two composite interpretations should not be reused. Add a separate erratum and preserve the original package. The machine-readable dictionary is also herbivore-only; the predator contract exists in prose and should receive an equivalent versioned dictionary before cross-role consumers depend on it. [4][5][6][9]

4 Upgrade design and balance

Stat Lines can make upgrade authoring more deliberate without reducing design to optimization. Their practical role is to show which tradeoff changed, where a benefit moved a cost, how variable the result is across seeds, and whether an effect survives a matched comparison. Designers still choose the desired play pattern and acceptable ecological consequences.

Step	Designer task	Evidence check
1 Intent	State the intended decision, benefit, cost, and species fantasy.	Define success and failure before the run.
2 Freeze	Author the upgrade, scenario, seed roles, timing, and comparison scope.	Capture ruleset and immutable launch snapshot.
3 Measure	Run baseline and variant on the same cases.	Use raw counts, validity states, distributions, and direct outcomes.
4 Interpret	Ask whether the effect is readable, viable, and ecologically acceptable.	Keep observed fact, inference, balance judgment, and player value separate.
5 Revise	Change data, costs, tradeoffs, or presentation and retest.	Create new evidence; do not rewrite the previous report.

COMPOSITE METRIC BOUNDARY

Metric	Current meaning	Useful for	Not approved for
RFS	Replication fitness score	Reading population change with birth conversion	Direct payout or universal fitness rank
APS	Actual prey score for herbivores	Summarizing applicable replication and survival components	Cross-role comparison or species verdict
AHS	Actual hunt score for predators	Summarizing hunt, replication, starvation, and crowding components	Pooling with APS without normalization
FPO	Final population in one window	Direct outcome and accounting reconciliation	Assuming that higher is always better

5 Predictive AI research

The research program asks whether deterministic simulation plus an auditable AI workflow can help a small team discover, test, explain, and promote better rules without surrendering human design authority. Its organization is intentionally evidence-first: Experiment, Run, Report, Analysis, and Human Decision are different artifacts with different owners. [7]

Phase	Status	What it establishes	Promotion boundary
P0	Complete	IDs, hypothesis framing, ownership, and decision protocol	Program can run bounded studies
P1	Complete within bounds	Reproducibility, fingerprints, replay, and known telemetry limits	Instrument trusted only within recorded coverage
P2	Complete within bounds	Evidence-backed diagnosis of a bounded ecology outcome	Explanation does not imply general cause
P3	Closed as bounded	Forecasts, order effects, thresholds, and human approval workflow	No generalized predictive or balance capability
P4	Not started	Translation into upgrade, event, explanation, or curator candidates	Needs a fresh human-owned contract
P5	Not started	Repeatability across scenarios and contributors	No automatic promotion from P4
P6	Not started	Decision on product, tooling, process, or external review	Production use remains a new gate

WHAT THE RESEARCH CAN PROVIDE

- Faster evidence: track time, manual steps, reruns, review burden, and decision rate.
- Better diagnosis: cite reports, run controlled follow-ups, and measure unsupported-cause errors.
- Earlier warnings: use held-out tests, range coverage, confidence, and out-of-distribution checks.
- Clearer player explanations: translate accepted evidence and validate comprehension in playtests.

CONDITIONS BEFORE THE NEXT PREDICTIVE STUDY

Before another confirmatory study, either close or explicitly accept the remaining evidence-hygiene gaps: versioned metric definitions, a sealed AI context manifest, a seed registry, event-level scoring, endpoint families, complete provenance, and raw-artifact retention. The roadmap adds a production gate: core loop, economy, and metric meaning must be stable enough to make the result useful. [1][7]

6 Overlap opportunity and risk

The strongest overlap is shared evidence and experimental machinery. The weakest overlap should remain decision authority. Currency and Predictive AI may consume the same immutable reports, but they answer different questions and should not share one opaque score. [8]

Dimension	Overlap	Shared value	Boundary
Metrics and provenance	High	One dictionary, seed, ruleset, window, and validity contract	Consumers do not recalculate formulas
Batch and comparison tooling	High	Matched panels support payout tests, balance tests, and forecasts	Each consumer keeps separate outputs
Hypothesis formation	Medium	Prediction misses can expose missing telemetry or unclear labels	AI proposals need human pre-approval
Player explanation	Medium	Accepted evidence can support readable cause and effect	Research scores stay out of player UI
Design authority	Low by design	Evidence informs the decision	Humans approve balance, payout, fun, and promotion

PRIMARY RISKS AND CONTROLS

Priority	Risk	Smallest control
P0	Metric semantic drift produces official-looking but wrong scoring.	Version definitions; add an EX-007 erratum; reject unknown versions.
P0	One composite becomes a payout, species verdict, or AI success score.	Show raw and per-stat outcomes first; keep composites secondary.
P0	Policy or evaluation writes back into run state or past evidence.	Run consumers after completion and save separate versioned outputs.
P1	APS and AHS are pooled across roles without normalization.	Keep role-specific contracts and require an explicit comparison design.
P1	A reference panel is overfit and treated as general balance.	Rotate held-out seeds and scenarios; label valid ranges and freshness.
P2	Research diagnostics leak into player UI as species ratings.	Use only human-approved explanations in player surfaces.

7 Combined operating model

The combined model uses one immutable evidence spine and several narrow consumers. It creates leverage because every new trustworthy report can support explanation, economy calibration, upgrade review, and future prediction without duplicating the simulation formulas.

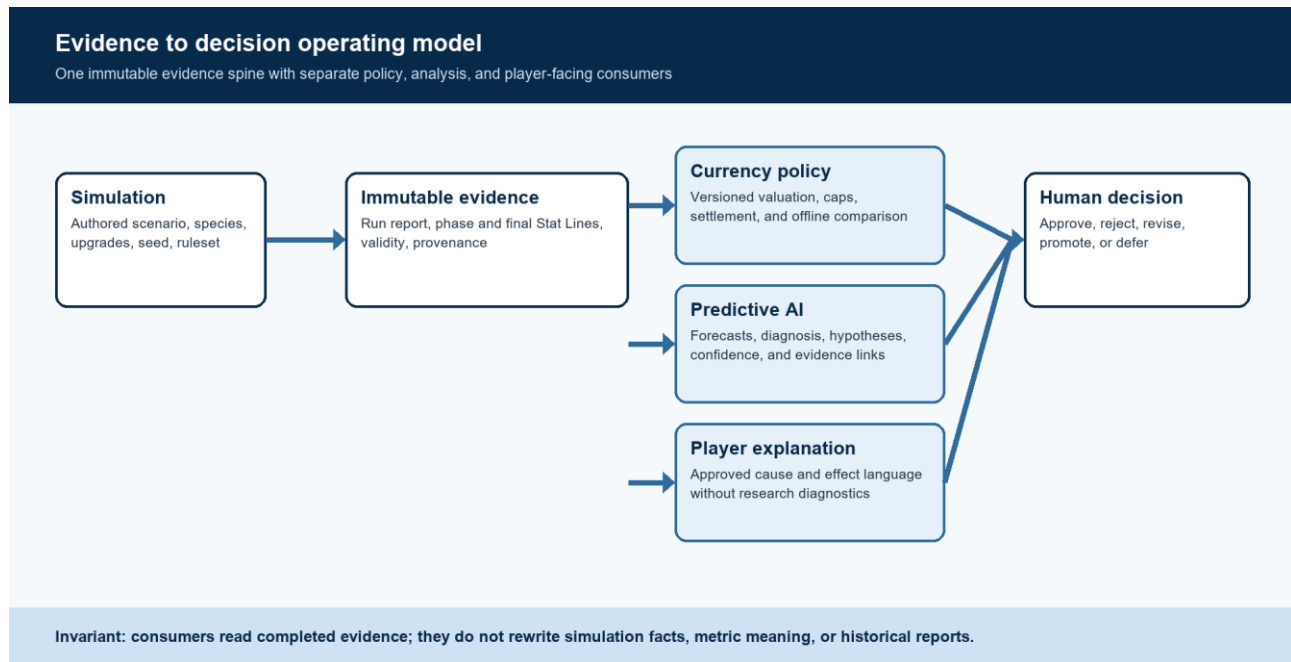


Figure 1 Evidence to decision operating model

RECOMMENDED DELIVERY SEQUENCE

When	Deliverable	Exit evidence
Now in S3	Freeze consumed metric meanings; draft currency-policy-v1; add predator dictionary and EX-007 erratum.	Every policy input has a stable ID, window, unit, status rule, and provenance.
S4	Approve a Forest Edge reference panel; compare payout candidates offline; run matched-seed upgrade reviews.	Human decision records select or reject candidate policies and upgrade changes.
S5	Validate player-facing cause and effect language.	A comprehension playtest can identify role, pressure, selected effect, and outcome cause.
S6	Implement versioned settlement and the first persistent Genome unlock.	One Hare unlock survives restart, toggles between runs, and applies only when active.
Later gate	Run one P4 question, then P5 or P6 only if separately approved.	Evidence shows workflow value beyond one fixture before production promotion.

8 Meeting decisions and evidence register

A useful meeting outcome is a small set of explicit approvals, not a final balance model. Record each decision with its scope and the evidence it authorizes.

Decision	Recommended disposition	Owner to confirm
Operating boundary	Approve measurement, valuation, prediction, and human review as separate layers.	Josh and Sim
Metric contract work	Approve the small S3 package and preserve historical reports with an erratum.	Sim for meaning; Josh for consumers
Currency status	Keep survivor-count rewards as prototype plumbing until offline calibration passes.	Josh with Sim review
Predictive AI timing	Keep F20 on hold during S3; decide after the contract gate whether one P4 question earns capacity.	Josh and Sim
Genome dependency	Do not make production Genome progression depend on unapproved payout or predictive scores.	Josh with Sim review

EVIDENCE BOUNDARY

This brief reflects the current ProjectMain working-tree planning and implementation state reviewed on 17 September 2026. It does not replace the roadmap, sprint control record, formula contracts, research plan, experiment packages, or human decision records. Where a historical report conflicts with the current versioned metric dictionary, the dictionary controls new work and the historical record remains unchanged with an added erratum.

Source register

No	Record	Used for
1	Production roadmap ROADMAP.md	Milestones, Sprint 4 to Sprint 7 horizon, feature status, and F20 hold.
2	Sprint 3 control record docs\Sprints\S3-control-record.md	Active scope, delivery order, Genome boundary, and no new predictive experiment.
3	Scientific Data Economy docs\SCIENTIFIC_DATA_ECONOMY.md	Approved direction, prototype survivor reward, settlement questions, and Genome dependencies.
4	Herbivore Stat Line contract docs\HERBIVORE_STATISTICS_VALIDATION.md	Formula, validity, export, evidence, and applicability boundaries.
5	Predator Stat Line contract docs\PREDATOR_STATISTICS_VALIDATION.md	Role-specific AHS formula and cross-role normalization warning.
6	Metric dictionary version 1 docs\Research\METRIC_DICTIONARY_V1.json	Machine-readable herbivore metric names, units, windows, directions, and statuses.
7	AI-Assisted Ecology Laboratory Research Plan docs\Research\AI_ASSISTED_ECOLOGY_LAB_RESEARCH_PLAN.md	Hypothesis, protocol, phase gates, benefits, limits, and next-selection conditions.
8	Stat Line and Predictive AI concern record docs\Planning Concerns\STATLINE_PREDICTIVE_AI_CROSSOVER.md	Active overlap guardrails, ownership, and first-pass recommendations.
9	EX-007 report and analysis docs\Research\Experiments\EX-007-Predictive-Statline-Interventions	Historical labels, measurement correction, direct outcomes, and bounded decision context.
10	Currency implementation path SimulationRunResult.cs and SpeciesSimulationPreview.cs	Current survivor-count currency assignment and phase or terminal application.